Team Involvement in Demonstration

Date	3 July 2026
Team ID	XXXXXX
Project Name	Credit Card Approval Prediction
Maximum Marks	1 Mark

Team Involvement in Demonstration:

Team Members & Contributions

#	Name	Role	Primary Contributions	Contribution %
1	**Sirisha Medisetti**	Team Lead / ML Engineer	Project coordination, target variable engineering, model training, Flask frontend (templates, CSS, JS), Render deployment, GitHub repository management	30%
2	**Sai Satya Sri Mutyala**	Backend Develop er	Flask route implementation, preprocess_input() func tion, prediction history logic, form-to-model integration, session management	20%
3	**Rekha Prasanna Nagabathul a**	Data Engineer / DevOps	Kaggle dataset acquisition, data cleaning, preprocessing pipeline (StandardScaler, LabelEncoder), Render deployment configuration (Procfile, render.yaml)	20%
4	**Hans Raj Mortha**	Data Analyst / QA	Exploratory Data Analysis (EDA notebook — 11 sections), feature distribution visualizations, functional testing (8 test cases), documentation review	15%
5	**Sravya Sri Nakkalapu**	ML Engineer	Feature engineering (AGE_YEARS, YEARS_EMPLOYED,	15%

#	Name	Role	Primary Contributions	Contribution %
	di		IS_UNEMPLOYED, INCOME_PER_MEMBER), XGBoost training with scale_pos_weight, model comparison evaluation, metadata.pkl generation	

Story Assignment by Team Member

Team Member	Stories Assigned
Sirisha Medisetti	S1, S3, S7, S11, S14, S15, S17, S18, S20
Sai Satya Sri Mutyala	S4, S8, S9, S10, S19
Rekha Prasanna Nagabathula	S2, S5, S12, S16
Hans Raj Mortha	S6, S13
Sravya Sri Nakkalapudi	S7 (shared), S8 (shared), S11 (shared)

Story IDs correspond to the sprint backlog in docs/4. Project Planning Phase/Project Planning.md

---

Team Reflection

The project was completed in 5 sprints over 5 weeks. The most challenging aspect was handling the extreme class imbalance in the original dataset (initially 98.5% Approved / 1.5% Rejected). The team solved this by:

1. Expanding the definition of "bad" status to include STATUS=1 (30-59 days past due)
2. Excluding applicants with only X-status (no evaluated credit history)
3. Applying class_weight='balanced' to all Scikit-learn models
4. Applying scale_pos_weight = neg/pos ratio to XGBoost

This brought the class balance to 87%/13% and Random Forest accuracy from ~65% to 81.91%.